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OPNsense [Running] - Oracle VirtualBox
File Machine View Input Devices Help
| Forums:      https://forum.opnsense.org/ |      :::      :::
| Code:        https://github.com/opnsense |      `:::~::~:
| Reddit:      https://reddit.com/r/opnsense |      `:::~::~:
|-----|-----|
*** OPNsense.internal: OPNsense 26.7 (amd64) ***

LAN (em1)      -> v4: 10.10.10.1/24
OPT1 (em2)     ->
OPT2 (em3)     ->
WAN (em0)      -> v4/DHCP4: 10.0.2.15/24
                v6/DHCP6: fd17:625c:f037:2:a00:27ff:fe61:acd0/64

HTTPS: SHA256 59 C4 11 C7 1D 4E CA 2E 32 7B 2E A7 27 E7 F5 61
                E7 73 54 57 D7 17 CE A8 8D 5F 34 FC E7 39 61 09

0) Logout                      7) Ping host
1) Assign interfaces           8) Shell
2) Set interface IP address    9) pfTop
3) Reset the root password     10) Firewall log
4) Reset to factory defaults   11) Reload all services
5) Power off system            12) Update from console
6) Reboot system               13) Restore a backup

Enter an option: █
```

p1-01-opnsense-interfaces.png.

Report: Setting Up the OPNsense Firewall

Objective

The goal was to install and configure an OPNsense firewall in Oracle VirtualBox to act as the gateway for the lab network.

Setup

OPNsense 26.7 (amd64) was installed as a virtual machine with the hostname OPNsense.internal. The interfaces were configured as follows:

Interface	Adapter	Address
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WAN	em0	10.0.2.15/24 (DHCP), with an IPv6 address via DHCP6
LAN	em1	10.10.10.1/24 (static)
OPT1	em2	Not configured
OPT2	em3	Not configured

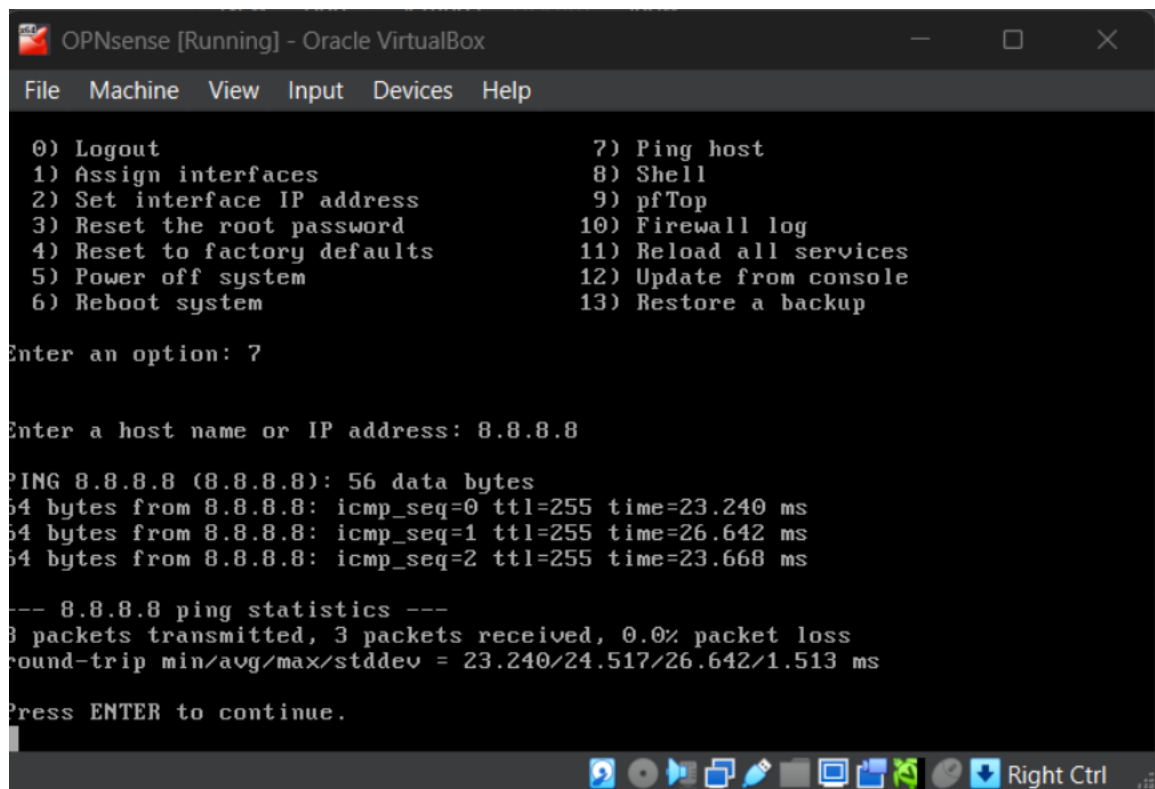
The WAN interface receives its address from VirtualBox, giving the firewall outside access. The LAN interface was set to 10.10.10.1/24, making it the gateway for the internal 10.10.10.0/24 network. OPNsense also generated an HTTPS certificate for its web interface, and its SHA256 fingerprint is displayed on the console.

Results

The firewall booted to the OPNsense console menu with both the WAN and LAN interfaces up and addressed correctly.

Conclusion

OPNsense was successfully set up with a working WAN connection and a LAN gateway at 10.10.10.1, ready for lab machines on the 10.10.10.0/24 network to connect to it.



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Enter an option: 7

Enter a host name or IP address: 8.8.8.8

PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=255 time=23.240 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=26.642 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=23.668 ms

--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 23.240/24.517/26.642/1.513 ms

Press ENTER to continue.
```

Internet Connectivity Test

Using the console's Ping host option (option 7), the firewall pinged 8.8.8.8 (Google's public DNS server). All 3 packets were received with 0% packet loss, and round-trip times ranged from 23.240 ms to 26.642 ms, averaging 24.517 ms. This confirms the WAN interface has working internet access.